Performance Testing

Date	23 October 2026
Team ID	0827CY231053
Project Name	Waypoint - AI Travel Itinerary Planner
Maximum Marks	5 Marks

Performance Testing

This worksheet defines a proposed benchmark plan. The repository contains manual functional test records, but no measured load-test report, CPU trace, memory trace or latency dataset.

Step 1: Testing Overview

Field	**Details**
Testing Tool Used	Proposed: k6 or Locust; FastAPI /docs and HTTP client for functional checks. No load tool run recorded.
Type of Testing	Planned baseline, load, stress and AI-generation latency checks.
Target Module / API	/api/health; /api/templates; authenticated /api/trips; trip generation and export.
Test Environment	Proposed isolated local SQLite baseline, then deployed Supabase environment. Record hardware and provider configuration before measurement.
Test Date	Not run; document prepared on 23 October 2026.

Step 2: Test Scenarios

S.No	**Test Scenario / Description**	**No. of Virtual Users**	**Duration (sec)**	**Expected Outcome**
1	Health and template read baseline	1	60	Routine API target under 2 s; no server errors.
2	Concurrent authenticated trip reads	10	120	Stable responses and user-scoped data.
3	Read-heavy stress ramp	10 to 50	180	Identify capacity and error onset without affecting real users.
4	AI generation vs local-rule generation	1 to 3	180	Record provider latency separately; confirm fallback on failure.

Performance Testing (continued)

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	Routine API < 2 s	Not measured	Not run	Target from requirements.
2	Response Time (Max)	< 5 s for routine API	Not measured	Not run	Proposed threshold; report p95 too.
3	Throughput (Req/sec)	Establish baseline	Not measured	Not run	Record successful requests / duration.
4	Error Rate	< 1%	Not measured	Not run	Separate server errors and expected validation errors.
5	CPU Utilization	< 80%	Not measured	Not run	Capture during steady load.
6	Memory Utilization	< 80%	Not measured	Not run	Record peak resident memory and host RAM.

Step 4: Observations & Analysis

- External AI latency and rate limits will differ from the local rules engine; measure each mode independently.

- SQLite and hosted Supabase have different concurrency and network characteristics.

- Capacity and uptime claims require measured evidence before being presented as achieved results.

Step 5: Screenshots / Evidence

No performance screenshots or benchmark output are available in the supplied files. Add the real tool report after executing the planned scenarios; no synthetic result is inserted.

Project sources: 2. Requirement Analysis/README.md; 6. Project Testing/README.md; backend main.py and AI engine.